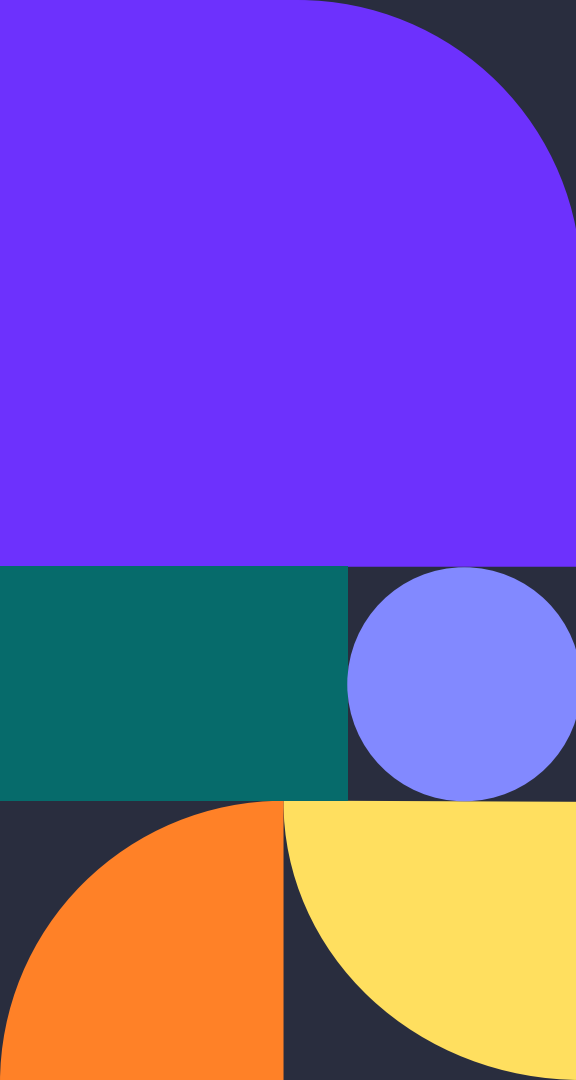




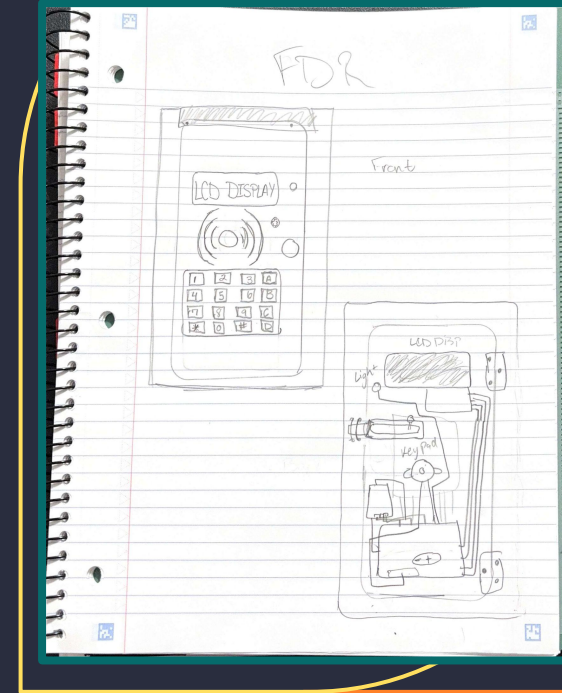
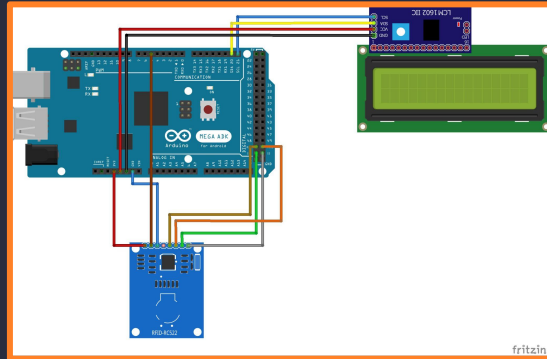
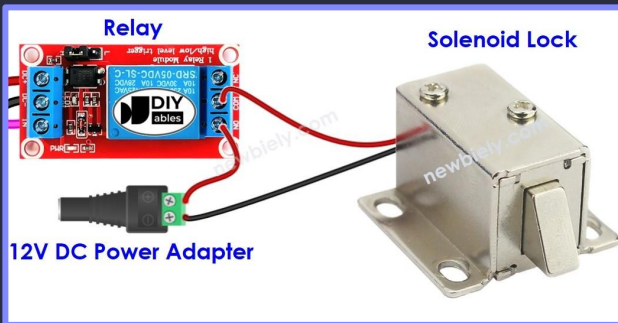
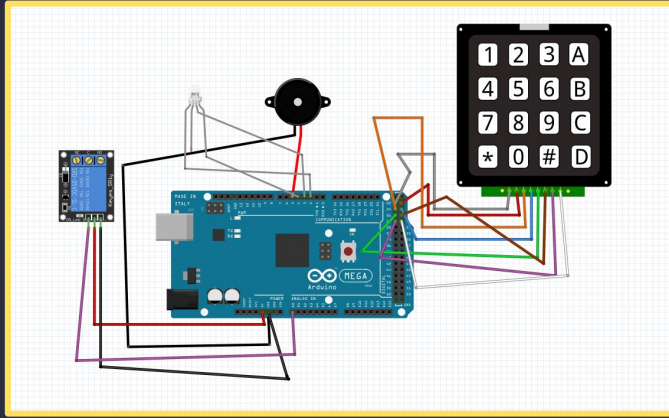
Group 1

Final Design Review (FDR)

Week 8

Armando H, Caleb M, Cesar O, Logan K, Collin E

Final Schematic/Drawing



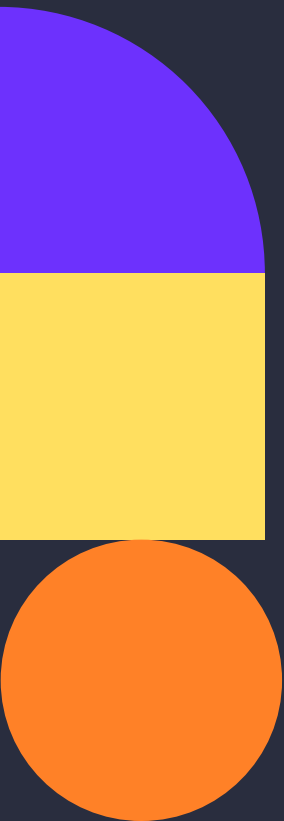
Code Architecture

The code is going to be run in the Arduino IDE, so the code will begin by setting up the hardware libraries. Next, the setup code for the hardware runs. The display will show a message. It'll ask for an input method, like an RFID tag or password. There will be code and a button used to record new passwords EEPROM stores passwords, fingerprints, etc.

After this setup code has run, then the main loop can start. For example, once an attempt has been made to enter the door, the device will check its memory to see if this matches any previously recorded records. If it does, the LED will be green and grant the user entry. If the user fails to enter a correct authentication, then the LED will be red and the buzzer will sound. There will be corresponding messages on the display as well, like access granted/denied. There will be code to change and remove passwords as well. After multiple failed attempts, the device won't allow the user to make another attempt at entry for a minute. After a successful attempt, the user needs to enter the door before it is locked again.

Bill of Materials (BOM)

- Arduino
- Fingerprint sensor
- 4x4 keypad
- RFID Reader and tags
- LCD display
- RGB LED
- Relay module
- Buzzer
- Resistors
- Jumper wires
- Solenoid lock
- Push button
- Potentiometer
- EEPROM
- Power Supply
- Circuit board
- Casing



Testing Plan

- Testing would involve making sure all three authentication methods worked properly.
- For the keypad, making sure all the inputs are read and show up on the display.
- For the RFID tags/reader we want to make sure the RFID's properly authenticate the user and make sure someone whose tag isn't properly authenticated isn't granted entry.
- For the fingerprint scanner, we want the person with the matching print to be able to get through the door.
- The device will be able to remember passwords, fingerprints, and tags.
- For all authentication methods, we want the user to not be able to brute force entry. To prevent this, the device will automatically lock down after several failed attempts. The door will also automatically lock shortly after a successful entry to prevent piggybacking.